Midterm Project — Simulation Study Report

Modules M01–M09 | Chapters 1–11 | Weight 20% | Difficulty ⋆⋆⋆

Modules: M01–M09 Chapters: 1–11 Weight: 20% Due: End of Week 14

Overview

The midterm project is an individual simulation study on a system of your choice. You will apply the full methodology covered through Module M09: conceptual modeling → implementation → input modeling → output analysis → scenario comparison.

This project demonstrates that you can execute a *complete* simulation study, not just run code.

1 System Selection

Choose **one** system from the list below, or propose your own (instructor approval required by the end of Week 12).

1. A university library help desk (peak vs. off-peak arrivals; single vs. two staff)
2. A fast-food drive-through (one or two service windows; time-varying demand)
3. A hospital outpatient lab (specimen processing; time-varying arrivals; multiple stations)
4. A warehouse order-picking operation (pickers, zones, batch vs. discrete picking)
5. A small-port ferry terminal (vessel interarrival, loading/unloading, berth allocation)
6. A data-center job queue (CPU-intensive vs. I/O-intensive jobs; priority scheduling)

If you propose your own system, the proposal must include: a 1-paragraph description, at least one non-trivial resource contention, and a plausible data source or rationale for synthetic parameters.

2 Required Components

2.1 1. Conceptual Model (20 pts)

- System description: boundaries, purpose, level of detail
- Entity–resource–event flow diagram
- State variable table

- Performance measures with units
- At least **six** stated assumptions, each with a brief justification

2.2 2. SimPy Implementation (25 pts)

- Clean, modular SimPy code
- Model parameters in a `dataclass` or configuration dictionary
- Random seeds passed as arguments (no global state)
- Verified against at least one analytical benchmark or manual trace

2.3 3. Input Modeling (15 pts)

- Identify and justify the distribution for each random input
- *Synthetic parameters* (no real data): state explicitly and justify plausibility
- *Provided or collected data*: apply MLE + at least one goodness-of-fit test

2.4 4. Output Analysis (20 pts)

- Determine the warm-up period (Welch's method or justified alternative)
- Run $\geq$ 30 independent replications; report 95% CIs for all key metrics
- Replication summary table: metric, mean, half-width, n

2.5 5. Scenario Comparison (15 pts)

- Define at least **three** meaningful scenarios (e.g., staffing levels, capacity)
- Use Common Random Numbers for paired comparisons where appropriate
- Report a CI for the difference between the best and baseline scenarios
- Identify and discuss the dominant factor driving system performance

2.6 6. Professional Report (5 pts)

- Executive summary ($\leq$ 150 words): system, question, recommendation, uncertainty
- All required sections present in the specified order
- All figures have labeled axes, captions, and text references
- Limitations section: at least 2 limitations that matter for the decision

3 Report Structure

Section	Content
1	Executive Summary ($\leq$ 150 words)
2	System Description & Scope
3	Conceptual Model
4	Input Modeling
5	Implementation Notes & Verification
6	Output Analysis: Warmup and Replications
7	Scenario Comparison & Results
8	Recommendation with Statistical Justification
9	Limitations and Future Work

Length: $\leq$ 12 pages (excluding figures and tables). Submit as PDF.

4 V&V Addendum

After submitting the report, you will complete the V&V addendum during Module M10. See `assignments/midterm_vv_adden`
The addendum applies the V&V checklist to your already-submitted model and is graded separately in M10.

5 Deliverables

Item	Format	Notes
Written report	PDF, $\leq$ 12 pages	See report structure above
Code	`.py` or `.ipynb`	Must run from repo root; include a `README` with run instructions
V&V addendum	PDF, 1–2 pages	Submitted separately in M10

Submit all three items via GitHub Classroom.

Grading Summary

Component	Points
Conceptual model	20
SimPy implementation	25
Input modeling	15
Output analysis (warmup + replications)	20
Scenario comparison	15
Professional report quality	5
Total	**100**

Common Mistakes to Avoid:

- **Single run**: reporting results from one replication is not a simulation study.
- **No warmup**: starting statistics from $t = 0$ in a steady-state simulation biases results upward.
- **Undocumented assumptions**: every simplification must be stated and justified explicitly.
- **Missing units**: all performance measures must include units (minutes, $/day, trucks/hour, ...).
- **Conclusions without uncertainty**: every recommendation must quote a CI or acknowledge estimation uncertainty.

Full rubric: `rubrics/midterm_rubric.pdf` | *Solutions and exemplar reports: the instructor package (available to instructors on request).*